25901866 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Minor Pathway Courses (MPC) - MG2MPCPHS100 - DEMOGRAPHY AND BIOSTATISTICS OF PUBLIC HEALTH

(2025 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. What is ageing population? [K] / [CO1]
2. Define Crude Birth Rate . [K] / [CO1]
3. Define sex ratio. [K] / [CO1]
4. What type of data does NFHS collect? [U] / [CO1]
5. What is a life table? [K] / [CO2]
6. Define population pyramid. [K] / [CO2]
7. What does MMR indicate? [U] / [CO2]
8. Differentiate between discrete and continuous data. [An] / [CO3]
9. Define mode with example. [U] / [CO3]
10. Mention two advantages of standard deviation. [K] / [CO3]

11. Write formula of Poisson distribution. [K] / [CO4]

12. Define NPV. [K] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

13. Describe the scope and significance of demography in public health. [U] / [CO1]

14. Define Census and explain its features. [U] / [CO1]

15. Discuss dependency ratio and demographic dividend. [U] / [CO2]

16. Explain age and sex composition. [U] / [CO2]

17. Define sampling and explain any two types of sampling methods. [U] / [CO3]

18. Differentiate between simple and complex tables with examples. [An] / [CO3]

19. Explain the assumptions of binomial distribution. [U] / [CO4]

20. Explain multiplication rule with example. [U] / [CO4]

21. Explain the properties of normal distribution. [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

22. Explain vital statistics and Civil Registration System in detail. [U] / [CO1]

23. Differentiate incidence and prevalence and explain their use in disease measurement. [An] / [CO2]

24. Define Biostatistics and discuss its objectives, importance, advantages and limitations in public health. [An] / [CO3]

25. Explain random variables and probability distributions in detail. [U] / [CO4]

[10x2 = 20]